

Preserving Multi-Modal Capabilities of Pre-trained VLMs for Improving Vision-Linguistic Compositionality



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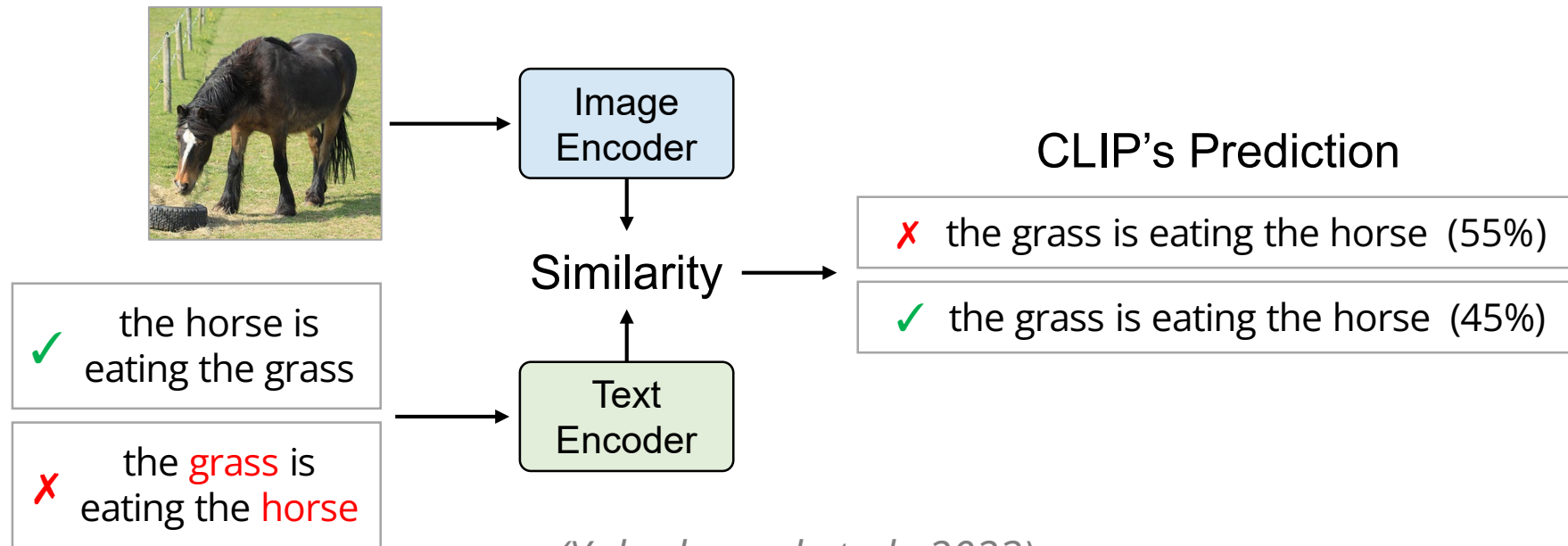
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Pre-trained VLMs: Successes and Challenges

VLMs like CLIP perform well in:

- Zero-shot Classification and Image-Text Retrieval.

However, they struggle with *compositional reasoning*.

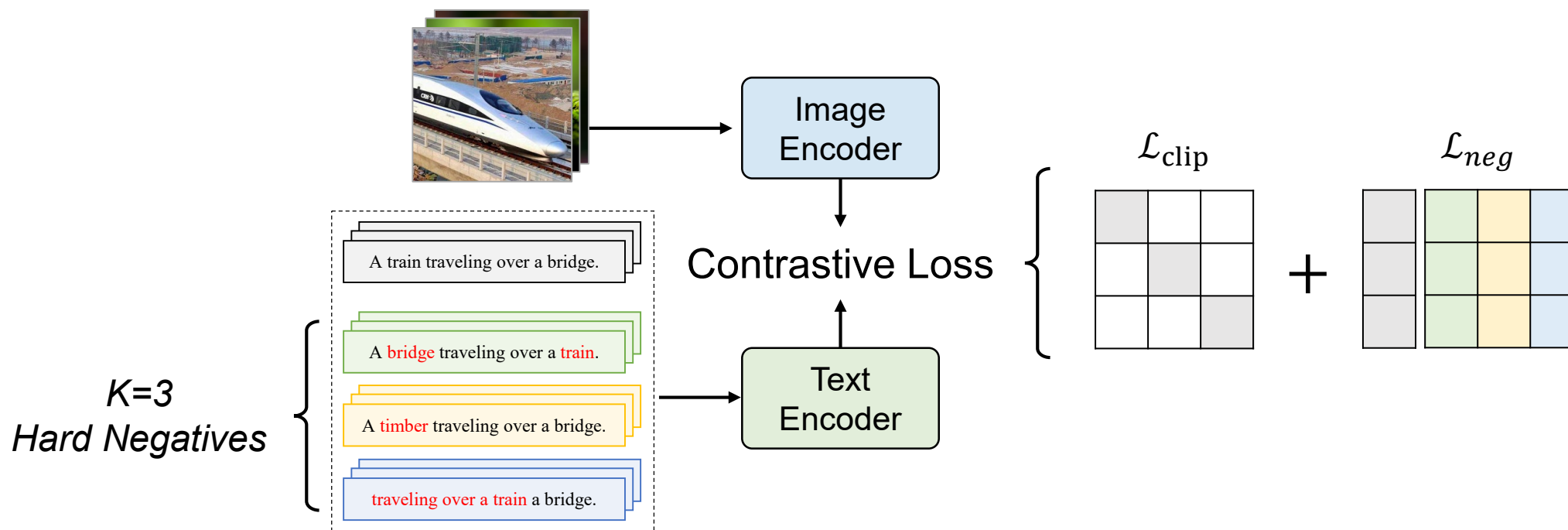


(Yuksekgonul et al., 2023)

Enhancing Compositional Reasoning in CLIP

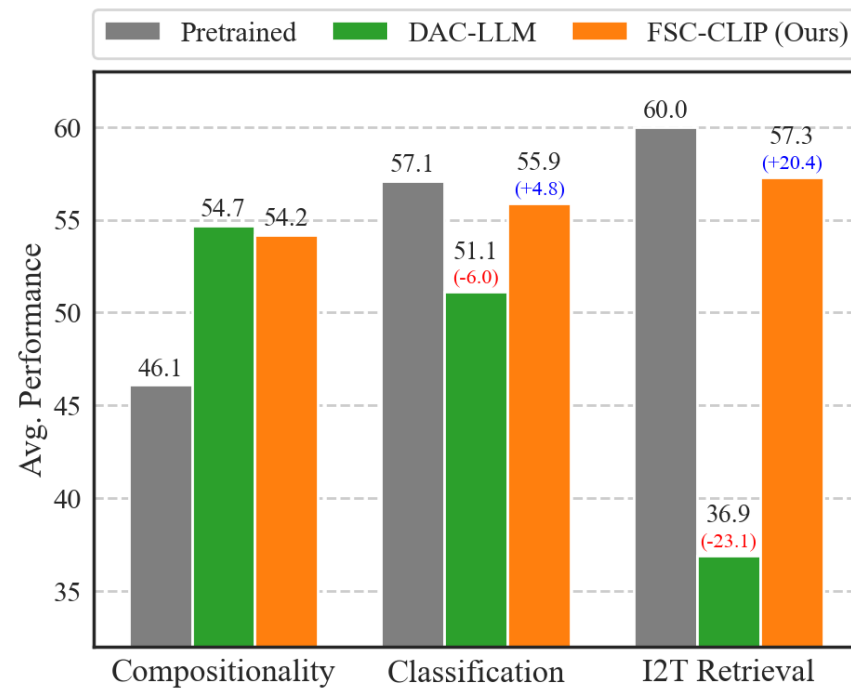
Fine-tuning CLIP with *hard negative captions* in contrastive loss:

- Specifically, NegCLIP (Yuksekgonul et al., 2023), DAC-LLM (Doveh et al., 2023), CE-CLIP (Zhang et al., 2024), and etc.



Trade-offs in Fine-tuning CLIP with Hard Negatives

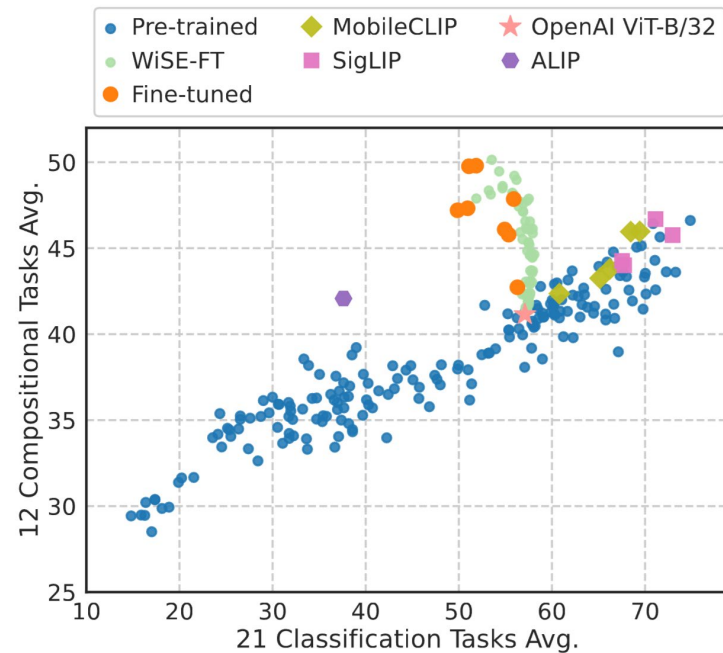
Fine-tuning with hard negatives (HNs) improves compositionality, but it *compromises* the model's original *multi-modal capabilities*.



Our work

Trade-offs in Fine-tuning CLIP with Hard Negatives

Fine-tuning with hard negatives (HNs) improves compositionality, but it *compromises* the model's original *multi-modal capabilities*.



[vl_compo](#) (Oh et al., 2024)

Correlation between zero-shot classification and compositional reasoning

Why Fine-tuning with HNs Degrades Multi-modal Performance?

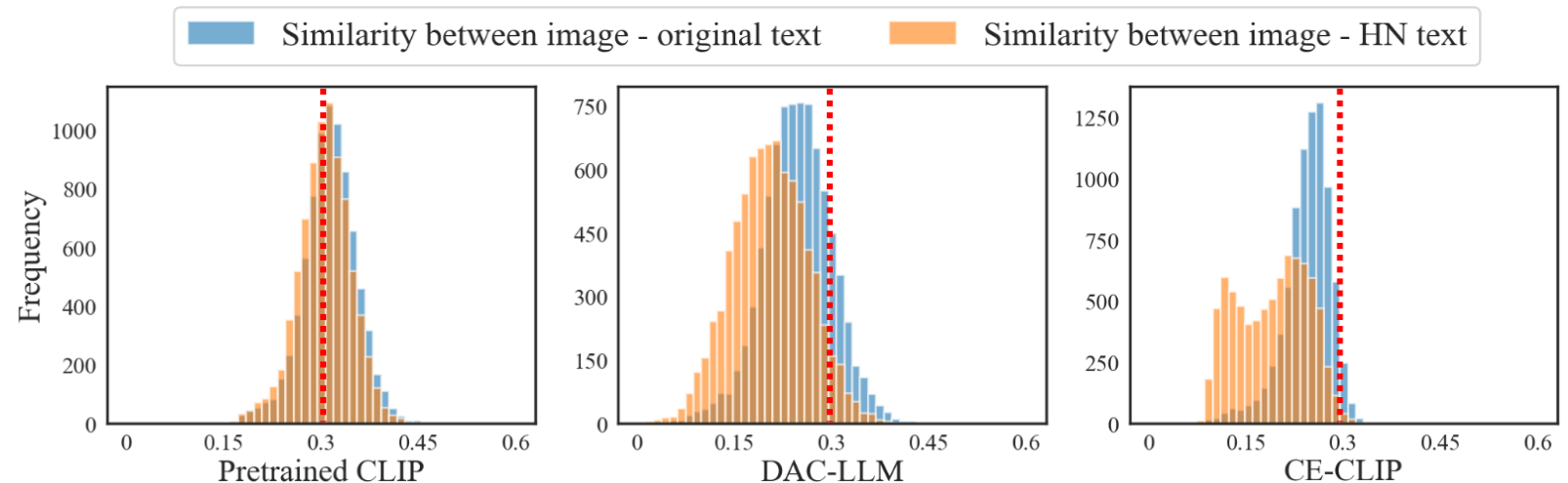
Pre-trained CLIP encodes original and HN texts *similarly*.

Fine-tuning with HNs **improves distinction**, but it **lowers overall similarity**.

LAION-COCO + HN captions
10,000 testing examples



- ✓ A glass bowl sitting on a table.
- ✗ A table sitting on a glass bowl.



Why Fine-tuning with HNs Degrades Multi-modal Performance?

Pre-trained CLIP encodes original and HN texts *similarly*.

Fine-tuning with HNs **improves distinction**, but it **lowers overall similarity**.

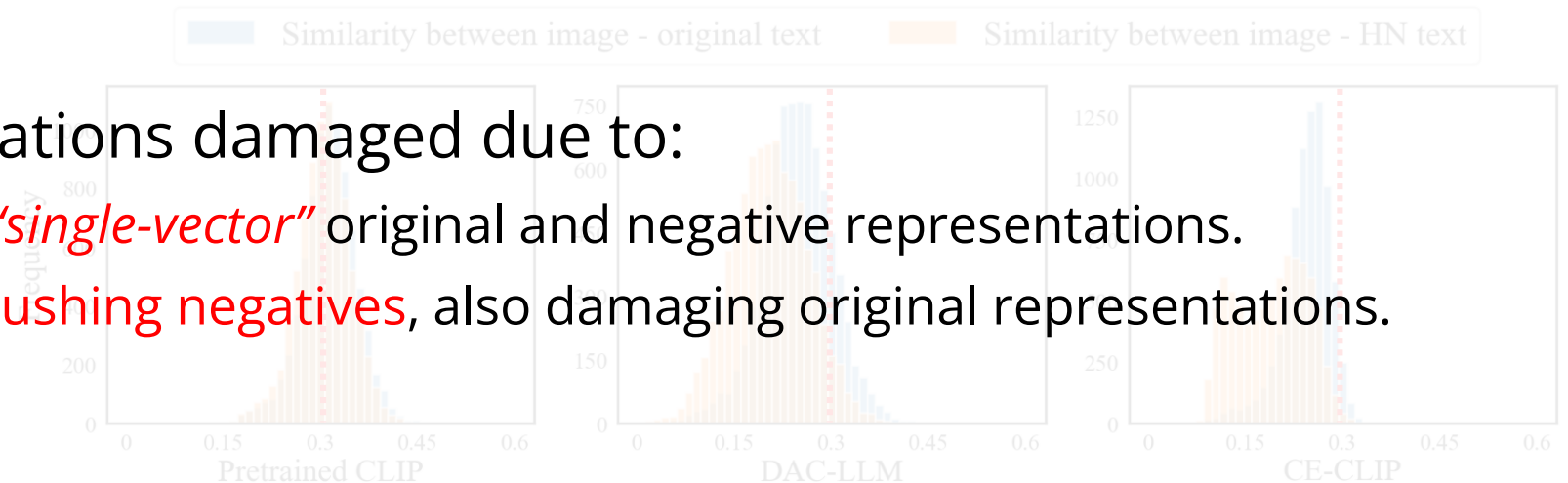
LAION-COCO + HN captions
10,000 testing examples

Pre-trained representations damaged due to:

- **Similar encoding** of “*single-vector*” original and negative representations.
- Hard negative loss **pushing negatives**, also damaging original representations.

✓ A glass bowl sitting on a table.

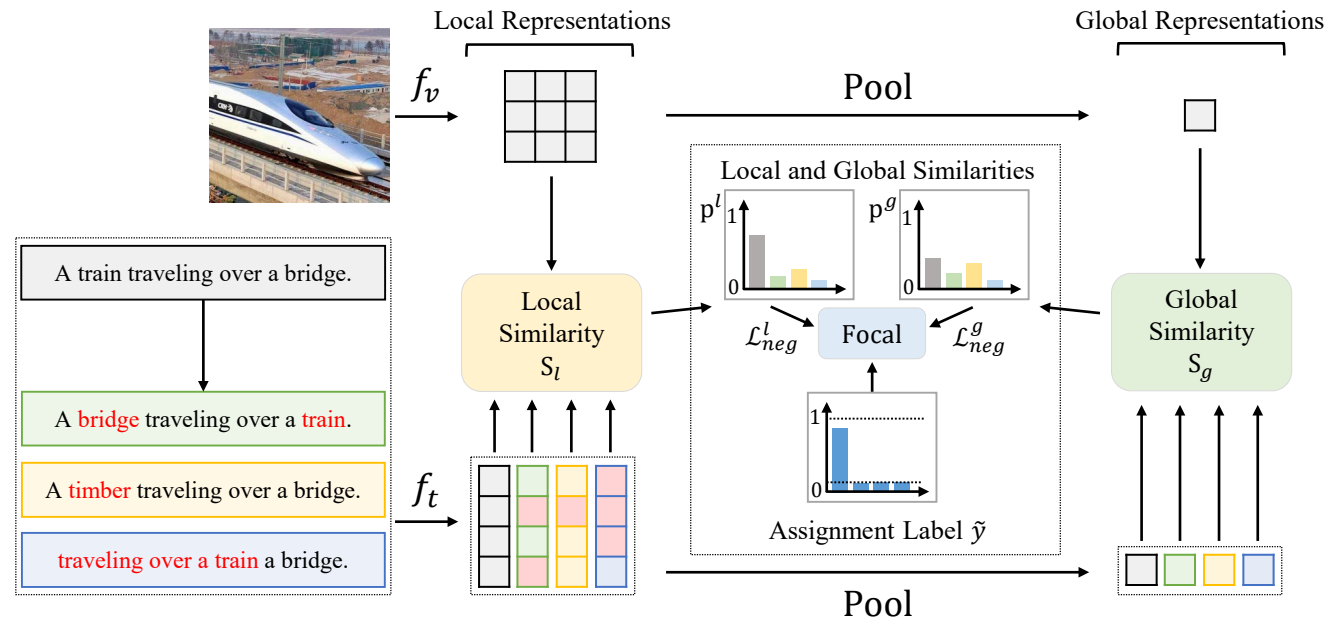
✗ A table sitting on a glass bowl.



FSC-CLIP: Preserving Multi-Modal Capabilities during Fine-Tuning

Improves *Hard Negative Supervision* by introducing:

- Fine-grained, Local Hard Negative (LHN) Loss
- Selective Calibrated Regularization (SCR)

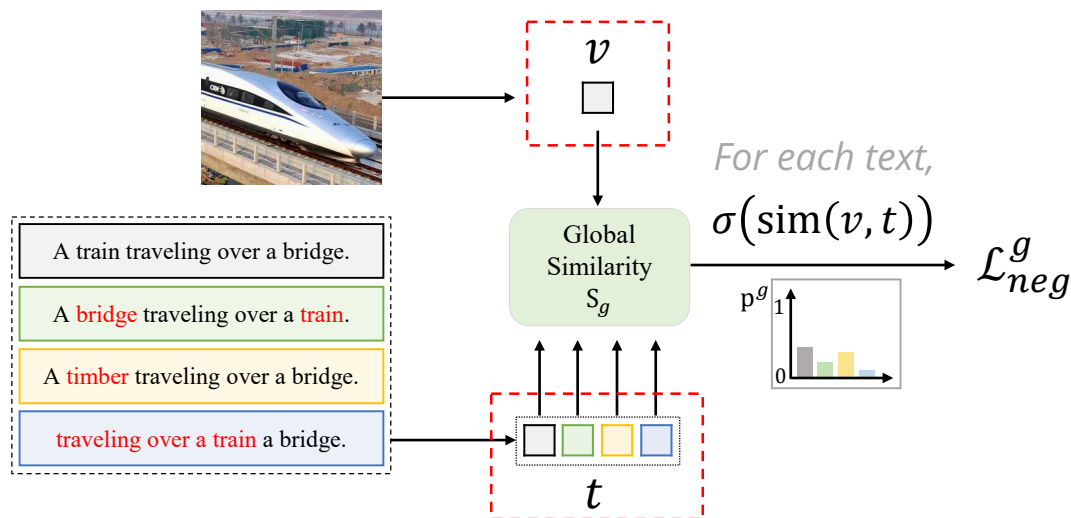


Motivation: Local Hard Negative (LHN) Loss

Single vector representations struggle with capturing *compositional language*.

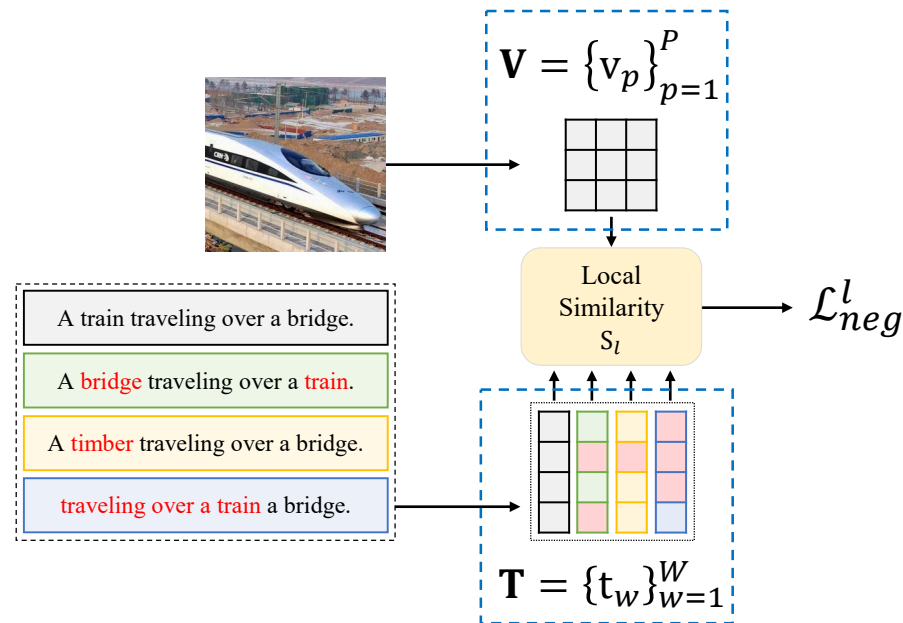
→ For image-text similarity, hard to differentiate original from HN texts:

$$S_g(I, T) = \text{sim}(v, t) = \exp(\cos(v, t)/\tau).$$



Local Similarity for the LHN Loss

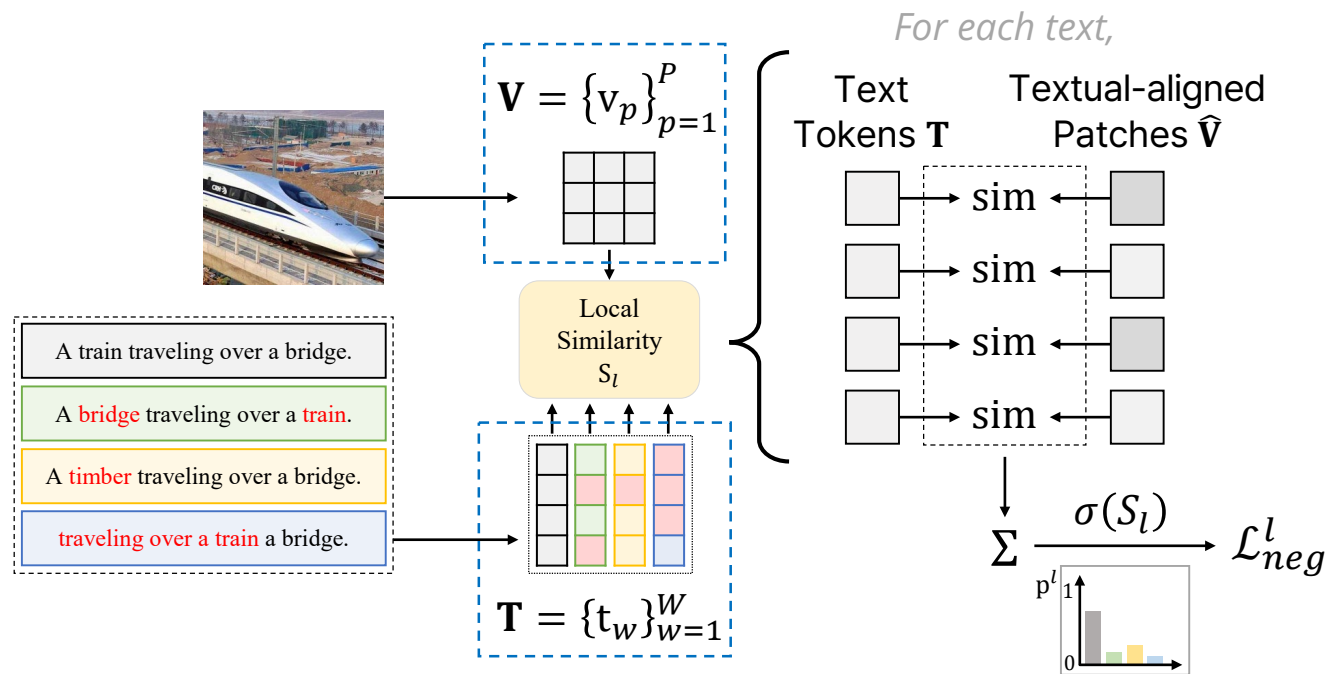
Leverage dense alignments between **patch** and **token representations**, for *fine-grained distinctions* between original and HN texts.



Local Similarity for the LHN Loss

Aggregate token-level similarities between the image and each text:

$$S_l(I, T) = \sum_{w=1}^W \text{sim}(\hat{v}_w, t_w) = \sum_{w=1}^W \exp(\cos(\hat{v}_w, t_w)/\tau).$$



Motivation: Selective Calibrated Regularization (SCR)

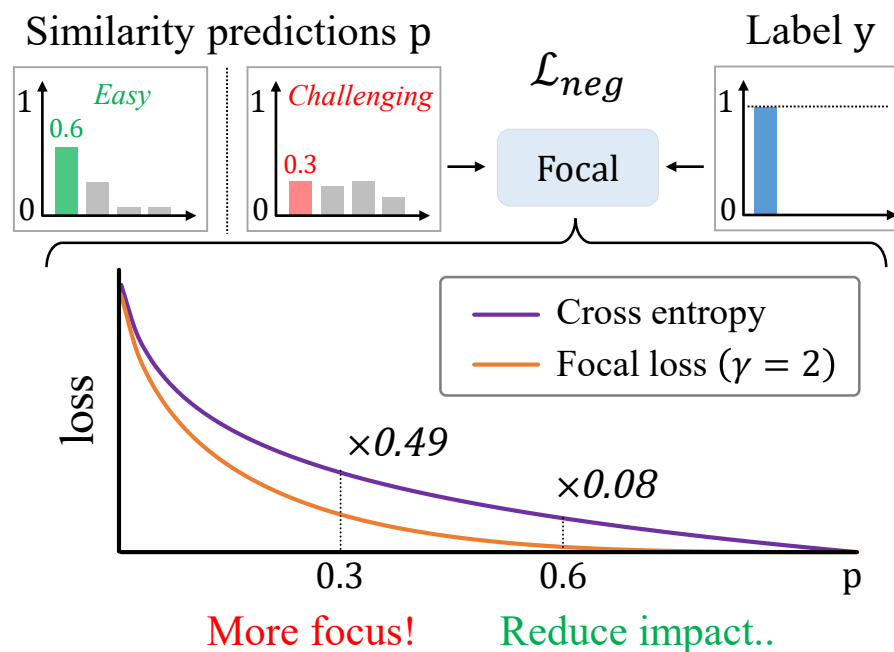
Hard Negative (HN) texts often have similar representations to the original.

To regularize Hard Negative losses,

- **Selective supervision** for *challenging* predictions
- **Calibrated positiveness** for hard negatives

Selective Calibrated Regularization (SCR)

Focus on *challenging* Hard Negative texts in $\mathcal{L}_{neg}$ $\rightarrow$ Focal loss



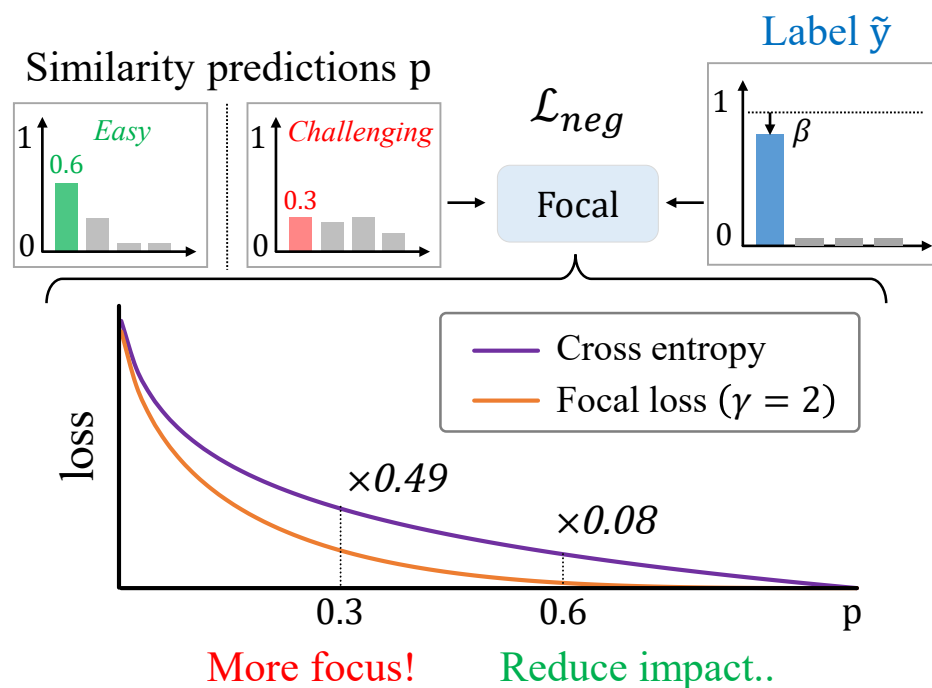
$$CE(p, y) = -\sum_k y_k \log(p_k) = \sum_k l_k$$

$$Focal(p, y) = \sum_k (1 - p_k)^\gamma \cdot l_k$$

Selective Calibrated Regularization (SCR)

Focus on *challenging* Hard Negative texts in $\mathcal{L}_{neg}$ → Focal loss

Potential *correctness* in HN texts → Label Smoothing



$$\tilde{y}_k = (1 - \beta) \cdot y_k + \frac{\beta}{1 + K}$$

$$\text{CE}(p, y) = -\sum_k y_k \log(p_k) = \sum_k l_k$$

$$\text{Focal}(p, y) = \sum_k (1 - p_k)^\gamma \cdot l_k$$

$$\text{Focal}(p, \tilde{y}) = \sum_k (1 - p_k)^\gamma \cdot \tilde{l}_k$$

FSC-CLIP: Overall Training Objective

Fine-tuning CLIP with hard negative texts:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{clip}} + \lambda_g \mathcal{L}_{\text{neg}}^g + \lambda_l \mathcal{L}_{\text{neg}}^l.$$

Selective Calibrated Regularization (SCR):

- Applied to both global and local HN losses, $\mathcal{L}_{\text{neg}}^g$ and $\mathcal{L}_{\text{neg}}^l$.

Evaluation:

- 11 Compositionality
- 21 Zero-shot Classification
- 3 Image-text Retrieval

Training:

- COCO (100K samples)
- CC-3M (100K samples)
- LAION-COCO (100K samples)

Method	LoRA	ARO	CREPE	EqBen	ImageCoDe	SugarCrepe	SVO Probes	VALSE	VL-Checklist	WhatsUp	Winoground	SPEC	Comp	ZS	I2T Ret	T2I Ret
CLIP (ViT-B/32)		57.5	23.8	26.5	21.7	73.1	84.1	67.5	70.8	41.5	8.8	31.9	46.1	57.1	60.0	45.8
<i>Fine-tuned: MS-COCO, 100K Samples</i>																
NegCLIP ¹		80.9	30.3	30.3	<u>26.4</u>	83.7	<u>90.8</u>	73.7	74.9	42.9	8.0	34.6	52.4	<u>55.9</u>	66.8	58.4
CE-CLIP ²		76.3	34.7	26.8	24.5	85.7	90.1	76.7	<u>76.9</u>	41.7	5.2	33.0	<u>52.0</u>	<u>49.9</u>	59.2	57.4
GNM-CLIP ³		57.1	17.4	28.3	25.0	78.7	89.2	71.1	70.6	<u>42.1</u>	10.2	33.1	<u>47.5</u>	56.3	66.1	55.5
MosaiCLIP ^{†,4}		82.6	-	-	-	-	90.7	-	76.8	-	-	-	-	-	-	-
NegCLIP [‡]		85.0	34.7	<u>29.8</u>	26.2	84.5	90.6	74.7	75.4	41.2	8.2	<u>34.2</u>	53.1	55.1	66.1	57.9
FSC-CLIP (Ours)		<u>85.1</u>	<u>42.2</u>	<u>29.8</u>	26.3	<u>85.1</u>	90.9	<u>75.3</u>	76.7	40.6	<u>9.5</u>	<u>34.2</u>	54.2	<u>55.7</u>	66.3	58.3
FSC-CLIP (Ours)	✓	85.2	42.9	29.7	26.5	82.1	90.4	75.0	77.2	41.7	6.0	33.2	<u>53.6</u>	<u>55.6</u>	65.3	57.2
<i>Fine-tuned: Conceptual Captions - 3M (CC-3M), 3M Samples</i>																
TSVLC ⁵ (RB)	✓	83.5	36.1	27.4	24.0	76.9	89.8	69.3	77.5	40.9	6.8	31.6	51.2	<u>54.9</u>	54.9	52.1
TSVLC ⁵ (RB+LLM)	✓	82.7	33.1	<u>27.6</u>	<u>24.6</u>	73.2	<u>89.7</u>	72.2	79.2	39.9	5.8	31.4	50.9	55.4	55.1	52.3
DAC-LLM ⁶	✓	<u>86.4</u>	<u>60.6</u>	25.6	22.8	85.3	88.9	70.5	<u>83.5</u>	<u>42.6</u>	4.8	30.8	<u>54.7</u>	51.1	36.9	52.4
DAC-SAM ⁶	✓	83.3	63.7	25.3	24.3	83.8	88.5	70.2	84.7	42.4	8.5	29.9	55.0	51.9	41.1	49.0
MosaiCLIP ^{†,4}		80.4	-	-	-	-	-	-	77.3	-	-	-	-	53.5	-	-
<i>Fine-tuned: Conceptual Captions - 3M (CC-3M), 100K Samples</i>																
NegCLIP [‡]		86.5	50.5	25.8	<u>24.6</u>	83.4	88.6	72.4	79.0	43.2	<u>7.0</u>	<u>32.8</u>	54.0	52.6	51.8	<u>54.1</u>
FSC-CLIP (Ours)		78.8	44.0	28.5	25.2	<u>84.3</u>	88.2	74.9	77.4	<u>42.6</u>	6.8	34.2	53.2	53.5	<u>55.8</u>	54.6
FSC-CLIP (Ours)	✓	84.4	50.6	27.7	24.5	82.3	88.8	<u>74.5</u>	80.3	42.1	5.0	32.2	53.9	53.6	56.1	54.0
<i>Fine-tuned: LAION-COCO, 600M Samples</i>																
CLoVe ⁷		83.0	41.7	26.9	25.3	84.6	87.9	71.8	66.6	41.8	6.5	31.7	51.6	51.0	53.1	56.0
<i>Fine-tuned: LAION-COCO, 100K Samples</i>																
NegCLIP [‡]		86.4	<u>48.7</u>	<u>27.2</u>	25.3	80.9	89.6	70.9	<u>76.0</u>	43.0	7.8	32.3	<u>53.5</u>	54.1	52.3	54.1
FSC-CLIP (Ours)		82.8	46.8	29.1	24.7	<u>82.6</u>	90.1	73.6	75.7	42.4	<u>6.8</u>	33.4	<u>53.5</u>	<u>55.3</u>	58.2	<u>55.5</u>
FSC-CLIP (Ours)	✓	<u>85.5</u>	54.4	29.1	<u>24.9</u>	80.6	<u>89.7</u>	<u>72.6</u>	78.4	<u>42.8</u>	5.8	<u>32.5</u>	54.2	55.9	<u>57.3</u>	54.3

[†]Numbers taken from the original paper. [‡]Our implementation, without additional image batch.

References: ¹(Yuksekgonul et al., 2023) ²(Zhang et al., 2024) ³(Sahin et al., 2024) ⁴(Singh et al., 2023) ^{5,6}(Doveh et al., 2022, 2023) ⁷(Castro et al., 2024)

Ablation Study

Global and Local HN losses, along with SCR, are all essential.

Method	Comp	ZS	I2T Ret	T2I Ret
CLIP (ViT-B/32)	46.1	57.1	60.0	45.8

id	$\mathcal{L}_{neg}^g$	$\mathcal{L}_{neg}^l$	Focal	LS	Comp	ZS	I2T Ret	T2I Ret
1	✓	-	-	-	54.0	53.6	47.4	53.7
2	-	✓	-	-	51.7	55.7	61.6	54.5
3	✓	✓	-	-	54.4	52.6	46.9	53.8
4	✓	✓	✓	-	54.2	54.2	53.1	54.8
5	✓	✓	-	✓	53.9	53.8	51.7	54.9
6	✓	✓	✓	✓	53.5	55.3	58.2	55.5
7	✓	-	✓	✓	52.8	55.3	57.1	55.6
8	-	✓	✓	✓	50.2	55.9	63.2	55.1

Fine-tuned: LAION-COCO 100K

Qualitative Results: Image to Text Retrieval

COCO-Counterfactuals → Compositional Reasoning + Retrieval

Image-Text Pair	Pre-trained CLIP	DAC-LLM	FSC-CLIP (Ours)
 GT: ☒ A table with some oranges and some apples.	Top-1: Two bowls of oranges are sitting on a metal surface. Top-2: A glass bowl filled with oranges on a table. Top-3: ☒ A table with some oranges and some cups.	☒ A table with some oranges and some cups. some oranges that are sitting on some wood A glass bowl filled with oranges on a table.	☒ A table with some oranges and some apples. ☒ A table with some oranges and some cups. A glass bowl filled with oranges on a table.
 GT: ☒ A man bending over a table with a lot of candles.	Top-1: ☒ A man bending over a cake with a lot of candles. Top-2: ☒ A man bending over a table with a lot of candles. Top-3: A man presents a cake with lit candles on it to a seated man.	a person holding a small cake with candles A person that is putting candles on a cake. A man blowing out candles on a cake	☒ A man bending over a table with a lot of candles. ☒ A man bending over a cake with a lot of candles. A man is being handed a birthday cake with lit candles.

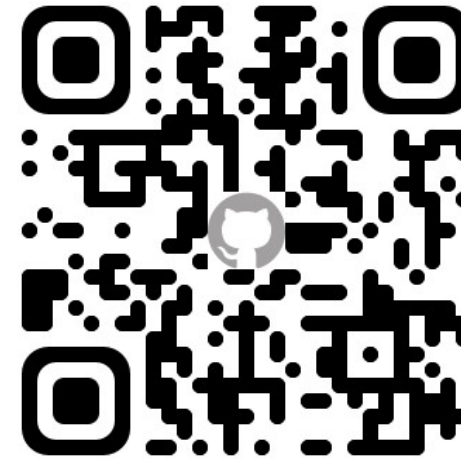
Takeaways

- Naive fine-tuning with hard negatives for compositionality harms multi-modal capabilities.
- *Local Hard Negative Loss* using fine-grained similarity, and *Selective Calibrated Regularization* controlling the hard negative losses help mitigate the issue.
- Training and evaluation codes, and model checkpoints are available!



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<https://www.linkedin.com/in/young-taek-oh/>



Project page and Codes:

<https://ytaek-oh.github.io/fsc-clip>

What is Next?

Compositionality in *Long Descriptions*.

SugarCrepe (Hsieh et al., 2023)



✓ A girl in a floral dress sitting next to a boy in a blue shirt eating food.

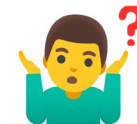
✗ A boy in a floral dress sitting next to a girl in a blue shirt eating food.

Single sentence!

DOCCI (Onoe et al., 2024)



A small pink figurine of a triceratops is placed on a bed of mulch in a blue plant pot. The triceratops is facing the left side of the image. Its head is tilted up toward the top left corner of the image as its front leg is stretched out in front of its body. Behind the triceratops in the plant pot is a green plant with white flowers, the plant extends directly up and out of view. The pot is placed on top of a grass area, behind the pot is a gray cement wall with a black metal wire fence in front of it in between the plant and the wall.



References

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(Sahin et al., 2024) Enhancing multimodal compositional reasoning of visual language models with generative negative mining, in WACV 2024.

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(Doveh et al., 2022) Teaching structured vision & language concepts to vision & language models, in CVPR 2022.

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(Hsieh et al., 2023) SugarCrepe: Fixing Hackable Benchmarks for Vision-Language Compositionality, in NeurIPS 2023.

(Onoe et al., 2024) DOCCI: Descriptions of Connected and Contrasting Images in EMNLP 2024.

Compositional Reasoning Benchmarks

Evaluate VLMs' ability to understand *compositional language* with images.

Test various types of relationships, through negative images/texts:

- Actions, relations and attributes in objects
- Spatial positions
- Linguistic phenomena
- etc.

Compositional Reasoning Benchmarks

Evaluate VLMs' ability to understand *compositional language* with images.

Test various types of relationships, through negative images/texts:

ARO (*Yuksekgonul et al., 2023*)

Relation



✓ the child is standing on the grass

✗ the grass is standing on the child

Attributes



✓ the large kettle and the metal pan

✗ the metal kettle and the large pan

Word Orders



✓ a man carrying a surfboard across the beach

✗ the beach a man carrying a surfboard across

Compositional Reasoning Benchmarks

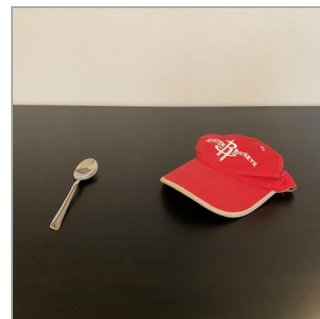
Evaluate VLMs' ability to understand *compositional language* with images.

Test various types of relationships, through negative images/texts:

WhatsUp (Kamath et al., 2023a)



- ✓ A scissors on a chair
- ✗ A scissors under a chair
- ✗ A scissors to the left of a chair
- ✗ A scissors to the right of a chair



- ✓ A cap to the right of a spoon
- ✗ A cap in front of a spoon
- ✗ A cap behind a spoon
- ✗ A cap to the left of a spoon

Spatial positions

Compositional Reasoning Benchmarks

Evaluate VLMs' ability to understand *compositional language* with images.

Test various types of relationships, through negative images/texts:

SVO Probes (*Hendricks et al., 2021*)

Query caption:
Couple walks by ocean.



Query caption:
A girl walking down the pier.



Query caption:
A boxer knocked onto the ground in a boxing ring.



Subject, Verb, Object

Compositional Reasoning Benchmarks

Evaluate VLMs' ability to understand *compositional language* with images.

Test various types of relationships, through negative images/texts:

Winoground (*Thrush et al., 2022*)



there is more **dirt** than **empty space** in the jar

there is more **empty space** than **dirt** in the jar

EqBen (*Wang et al., 2023*)



there are **3** bath towels in the scene

there are **4** bath towels in the scene

Counter-factual image-text pairs

Compositional Reasoning Benchmarks

Dataset	Testing Modality	Types of Negatives
ARO	Image → Text	Relation, Attribute, and Word Orders
CREPE		Atomic Foils, Negation, and Word Swaps
SugarCrepe		Add, Replace, and Swap Words
VALSE		Action, Coreference, Counting, Plurals, and etc.
VL-Checklist		Object, Attribute, and Relation
WhatsUp	Text → Image	Spatial Positions
ImageCoDE		Similar images and video frames
SVO Probes	Image → Text, Text → Image	Subject, Verb, and Object
Winoground		Commonsense
EqBen		Location, Counting, Attribute, and etc.
SPEC		Size, Position, Existence, and Count

Summary of compositionality datasets in our work for comparison.

Hard Negative Generation during Fine-tuning

Rule-based approach applied to every samples at each iteration.

Image-Text Pair	negclip	replace	bi-gram shuffle
 Three statues of elephants on the steps in front of an old building.	Three statues of steps on the elephants in front of an old building. Three statues of elephants on the steps in building of an old front. Three elephants of statues on the steps in front of an old building.	Three statues of pikas on the steps in front of an old building. Three statues of elephants into the steps in front of an old building. Three statues of megatherian mammal on the steps in front of an old building.	on the an old steps in building. Three statues front of of elephants Three statues building. of elephants an old steps in front of on the steps in on the front of an old building. Three statues of elephants
 Four different types of sandals with laces.	Four different sandals of types with laces. Four different laces of sandals with types. Four different types of laces with sandals.	Four different types of slingbacks with laces. Four inactive types of sandals with laces. Four different types of sandals with arms.	Four different laces. types of sandals with sandals with types of Four different laces. laces. types of Four different sandals with
 The small blue van is parked in front of a fence.	The blue small van is parked in front of a fence. The small blue van is parked in fence of a front.	The small blue regiment is parked in front of a fence. The small large van is parked in front of a fence. The small average van is parked in front of a fence.	is parked of a The small in front fence. blue van blue van in front of a is parked fence. The small The small in front blue van fence. is parked of a